

CARSON SLATER

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EDUCATION

PhD Statistical Science , Baylor University Advised by Amanda S. Hering . Research involves developing unsupervised learning methods for time series with applications in water distribution. Concentrations in <i>Data Science</i> and <i>Biostatistics</i>	2024 - 2027 [Expected]
M.S. Statistical Science , Baylor University (GPA: 3.86/4.0)	2023 - 2024
B.S. Applied Mathematics with Statistics , Wheaton College (IL) B.A. Economics (GPA: 3.46/4.0), <i>Edman Presidential Scholarship</i> <i>Varsity Baseball Athlete (NCAA-DIII)</i>	2019 - 2023

EXPERIENCE

Research Assistant Baylor University	January 2024 - Present Waco, TX
- Developing statistical methods for assessing urban water consumption with IoT meters. Research is funded by Denver Water. - Extracting, transforming, and loading unstructured data for analyses for Denver Water collaborators. - Collaborating with domain experts in urban water demand planning to assess water conservation interventions, implementing rigorous hypothesis testing and building Shiny dashboards to visualize individual and aggregate customer trends. - Engineering features from household-level billing and water usage time series to perform market segmentation for Denver Water's single-family residential customer base, coming up with seven user personas to characterize their population (~200k homes). - Edited and reviewed material for an undergraduate intro to data science course. (DOI: 10.7910/DVN/BZJW2E)	
Statistical Consultant Baylor University	August 2023 - December 2025 Waco, TX
- Consulted for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association. Published academic projects in peer-reviewed journals. - Delivered quality reports for each client and had regular meetings to discuss the project, the methodology and results.	
Academic Coach Wheaton Academy Center for Lifelong Learning	January - May 2023 West Chicago, IL
Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.	
Mathematics Teaching Assistant Wheaton College	August 2022 - May 2023 Wheaton, IL
Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.	

SKILLS

Technology	Docker, Positron ($\approx$ VS Code), RStudio, Git/Github, reveal.js, SQL, duckDB, $\LaTeX$
Python	pandas, numpy, PyTorch (Lightning), plotnine, seaborn, plotly, scikit-learn, Jupyter
R	tidyverse (dplyr, ggplot2, etc.), TensorFlow (Keras), torch, shiny, RMarkdown/Quarto, rstan, rjags, caret, tidy-
Stats & ML	models, targets (gittargets, proffer, etc.) statistical modeling, hypothesis testing, mixture models, time series, regression, linear mixed models, clustering, classification, deep learning, PCA, random forests, ensemble models (e.g., XGBoost), SVM, Bayesian methods, spatiotemporal modeling, convolutional neural networks, denoising diffusion probabilistic models

LEADERSHIP EXPERIENCE

President Baylor University Statistical Science Paper Club	May 2026 - Present Waco, TX
Sourcing and facilitating discussions on frontier papers in statistics, biostatistics, and machine learning for PhD students in the Statistics.	
President Baylor University Sports Analytics Club	March 2024 - May 2025 Waco, TX
- Led a team of undergraduates in submitting a March Madness bracket challenge submission on Kaggle. - Taught undergraduates basic version control, R programming, statistical and machine learning topics.	